

1. Project Overview

Project Title

Web Application for Drone Route Planning and Simulation

Project Goal

The goal of this project is to design and implement a web-based system that allows users to plan and simulate drone routes using a 2D map interface. The application integrates real-world map data and path planning algorithms to generate safe routes while avoiding obstacles.

Main Features

The system will allow users to:

- select a start point on a map
- select a destination point
- automatically load obstacles from map data
- add custom restricted zones
- generate an optimized route
- simulate drone movement along the route
- display drone altitude
- send mission commands such as **return home** and **land**

Key Focus

The primary focus of the project is the **route planning algorithm** and system architecture rather than complex visualization.

Map Environment

The system uses real geographic data through OpenStreetMap and extracts obstacle data such as buildings from the map.

Project Limitations

The first version of the system will support **one drone mission** at a time. However, the system architecture will be designed so that multiple drones can be supported in the future.